AI Capital versus Regular Capital: Real Interest Rates and Wages after Jumps, Faster Growth, and Anticipated Innovation

Research note (draft)

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Abstract

This note compares the dynamics of the real interest rate and the real wage after technical progress in AI capital and after progress in regular capital. Regular capital and labor form a Cobb-Douglas bundle, and AI capital is a substitute for that bundle. Progress is either disembodied (in the production function) or embodied (a falling price of new capital in the accumulation equation). Three experiments are considered: (1) a surprise permanent jump in technology; (2) a surprise permanent increase in its growth rate; and (3) an S-curve innovation centered on year 10 and known at year 0. Each experiment is solved under standard linear capital accumulation and under the moderate adjustment costs of Kiley (2004). Balanced growth paths are in closed form. For the same long-run output gain, AI progress raises the wage by $1/[1+(1-a)(\varepsilon-1)]$ of the output gain (74 percent here), while regular-capital progress raises it one-for-one. Embodied progress delays wage gains. With linear accumulation it initially *lowers* the real interest rate when it is ongoing or anticipated, because firms wait for cheaper capital. Adjustment costs largely remove that effect. Permanently faster AI progress has no balanced growth path when AI is a substitute.

1 Model

Notation follows Kiley (2000). Production nests a Cobb-Douglas bundle of regular capital and labor inside a CES aggregate with AI capital:

$$X = (\Gamma_K K)^a (ZN)^{1-a}, \qquad Y = \left[\omega^{1/\varepsilon} X^{\frac{\varepsilon-1}{\varepsilon}} + (1-\omega)^{1/\varepsilon} (\Gamma_A A)^{\frac{\varepsilon-1}{\varepsilon}}\right]^{\frac{\varepsilon}{\varepsilon-1}}, \qquad \varepsilon = 1.5. \tag{1}$$

AI substitutes for the bundle, and therefore for both labor and regular capital. Capital $j \in \{K, A\}$ is measured in efficiency units, and a new unit costs P_j in output:

$$K_j' = (1-d_j)K_j + I_j/P_j. \tag{2}$$

Disembodied progress raises Γ_j, and with it the productivity of the installed stock. *Embodied* progress lowers P_j, which makes new capital cheaper but leaves existing units unchanged. Two accumulation technologies are compared:

- *Linear*: no adjustment costs.

- *Adjustment costs*: $P_j \frac{c_j}{2}(i_j - x_j)^2 K_j$, with $i_j = I_j/(P_j K_j)$. The cost vanishes at the balanced-growth investment rate x_j, and $c_j = 1/x_j$, so that the elasticity of i_j with respect to Q_j/P_j is one (the paper's "moderate" case).

Utility is logarithmic and hours are fixed. Calibration: $a = 0.282$, $d_K = 0.066$, $d_A = 0.20$, trend growth $g_Z = 1.5$ percent, initial $r = 7.5$ percent and an initial AI income share of 3 percent. Because $\varepsilon > 1$, the AI share rises as AI becomes cheaper.

Proposition 1 (Balanced growth path)**.** *Let Γ_j grow at $\gamma_{\Gamma j}$ and P_j fall at γ_{qj}, and write $G_j = (1+\gamma_{\Gamma j})(1+\gamma_{qj})$. A balanced growth path exists if $\varepsilon = 1$, or if $\varepsilon \neq 1$ and $G_A = 1$. Output grows at*

$$\ln(1+g) = \ln(1+g_Z) + \frac{a\omega \ln G_K + (1-\omega)\ln G_A}{(1-a)\,\omega} \quad (\varepsilon = 1),$$
$$\ln(1+g) = \ln(1+g_Z) + \frac{a}{1-a}\ln G_K \quad (\varepsilon \neq 1).$$

With $1+r = (1+g)/B$, user costs $u_j = (1+r)(1+\gamma_{qj}) - (1-d_j)$ and effective costs $\lambda_j = P_j u_j/\Gamma_j$:

$$s_A = (1-\omega)\lambda_A^{1-\varepsilon}, \qquad \textit{labor share} = (1-a)(1-s_A), \qquad P_X^{1-\varepsilon} = \frac{1-s_A}{\omega},$$
$$\ln\frac{W}{Z} = \frac{1}{1-a}\ln P_X - \frac{a}{1-a}\ln\lambda_K + \textit{const}, \qquad \ln\frac{Y}{Z} = \ln\frac{W}{Z} - \ln(1-s_A) + \textit{const}.$$

Adjustment costs are zero on the path, $Q_j = P_j$, and $P_K K/Y = a(1-s_A)/u_K$, $P_A A/Y = s_A/u_A$.

The embodied user cost includes an obsolescence term: $u_j \approx r + d_j + \gamma_{qj}$.

Proposition 2 (Level shifts with the same long-run output gain γ_Y)**.** *Disembodied and embodied progress have the same long-run effects when $\ln\Gamma_j = -\ln P_j$. The long-run interest rate is unchanged.*

1. Regular capital*: the AI share and labor share are unchanged, so $\Delta\ln W = \gamma_Y$. The required shock is $\ln(\Gamma_K/P_K) = \frac{1-a}{a}\gamma_Y$.*
2. AI capital*: only s_A moves, and*

$$\Delta\ln W = \frac{\gamma_Y}{1+(1-a)(\varepsilon-1)}, \qquad \Delta\ln(\textit{labor share}) = \Delta\ln(1-s_A) = -\frac{(1-a)(\varepsilon-1)}{1+(1-a)(\varepsilon-1)}\,\gamma_Y.$$

Proof. From Proposition 1, $\ln(Y/Z) = \frac{1}{(1-a)(1-\varepsilon)}\ln(1-s_A) - \ln(1-s_A) - \frac{a}{1-a}\ln\lambda_K + \text{const}$. Regular-capital progress moves only λ_K. AI progress moves only s_A, and $\ln W = \ln Y + \ln(1-s_A)$. □

With $\gamma_Y = 10$ percent, regular-capital progress raises the wage 10 percent and AI progress raises it 7.4 percent. The AI share rises from 3.0 to 5.5 percent, and the labor share falls from 69.6 to 67.8 percent. Because AI is small, the same output gain requires Γ_A/P_A to rise 3.4-fold, against a 29 percent rise in Γ_K/P_K.

Proposition 3 (Faster growth)**.** Regular capital*: a permanent increase in $\gamma_{\Gamma K}$ or γ_{qK} moves the economy to a new balanced growth path with $\Delta\ln(1+g) = \frac{a}{1-a}\ln G_K$ and $\Delta\ln(1+r) = \Delta\ln(1+g)$, for any ε. The wage grows at the new rate g, and its level relative to Z falls because λ_K rises. The fall is larger for embodied progress, through the obsolescence term in u_K.* AI capital with $\varepsilon > 1$*:*

there is no balanced growth path. $\lambda_A \to 0$, the AI share $\to 1$, the labor share $\to 0$, and growth and the interest rate rise without bound. This is the explosive limit derived in the companion note. AI capital with $\varepsilon = 1$*: there is a balanced growth path with* $\Delta \ln(1+g) = \frac{1-\omega}{(1-a)\omega} \ln G_A$*. Delivering the same growth increase as regular-capital progress then requires AI progress about* $(1-a)\omega/(1-\omega) \approx 23$ *times as fast as the increase in output growth.*

Two equivalences organize the results. With a Cobb-Douglas bundle, disembodied regular-capital progress is the same as labor-augmenting progress. With $\varepsilon = 1$, disembodied progress in either capital is the same as Hicks-neutral TFP growth. Its paths are therefore identical whichever capital it attaches to. In both cases only *embodied* progress distinguishes one kind of capital from another.

2 Experiments

Every experiment is announced, or occurs, at $t = 0$ with the economy on its initial balanced growth path. Each is solved as a nonlinear perfect-foresight problem (Newton's method, 250 years, residuals below 10^{-10}). Wages are log deviations from the pre-shock path (W/Z on the initial trend). The labor share is relative to gross output.

Case 1: surprise permanent jumps

Table 1: Case 1: surprise permanent jump, same 10% long-run output gain. The shock is in log points; r_1 is the rate between years 0 and 1.

		shock	r_1 (pct.)	peak r (pct.)	year	$\ln W$, t=10 (pct.)	$\ln W$, long run (pct.)	labor share long run (pct.)
K disemb.	linear	0.255	8.38	8.38	0	9.4	10.0	69.6
K disemb.	adj. costs	0.255	7.69	7.69	0	8.4	10.0	69.6
K emb.	linear	0.255	10.64	10.64	0	8.2	10.0	69.6
K emb.	adj. costs	0.255	8.14	8.14	0	4.4	10.0	69.6
AI disemb.	linear	1.223	8.57	8.57	0	6.6	7.4	67.8
AI disemb.	adj. costs	1.223	8.05	8.05	0	5.5	7.4	67.8
AI emb.	linear	1.223	9.03	9.03	0	6.3	7.4	67.8
AI emb.	adj. costs	1.223	8.73	8.73	0	4.5	7.4	67.8

1. *Interest rates jump on impact and decay.* Embodied progress raises r more than disembodied progress, because it requires an investment boom before any output gain. Under linear accumulation, a jump in regular-capital technology raises r_1 to 10.6 percent (embodied) or 8.4 percent (disembodied). For AI the figures are 9.0 and 8.6 percent. Adjustment costs damp the regular-capital response (8.1 and 7.7 percent) more than the AI response (8.7 and 8.05 percent), because AI's adjustment cost per unit of investment is smaller.

2. *Wages.* Disembodied progress raises wages on impact, by 7.0 percent for regular capital and 3.0 percent for AI. Embodied progress raises them only as new capital is installed. With adjustment costs, after ten years wages are still 5.6 percentage points short of their long-run gain for regular capital, and 2.9 points short for AI.

3. *Labor share.* It is unchanged for regular capital. For AI it falls by 1.8 points, at once under disembodied progress and over 10–20 years under embodied progress with adjustment costs.

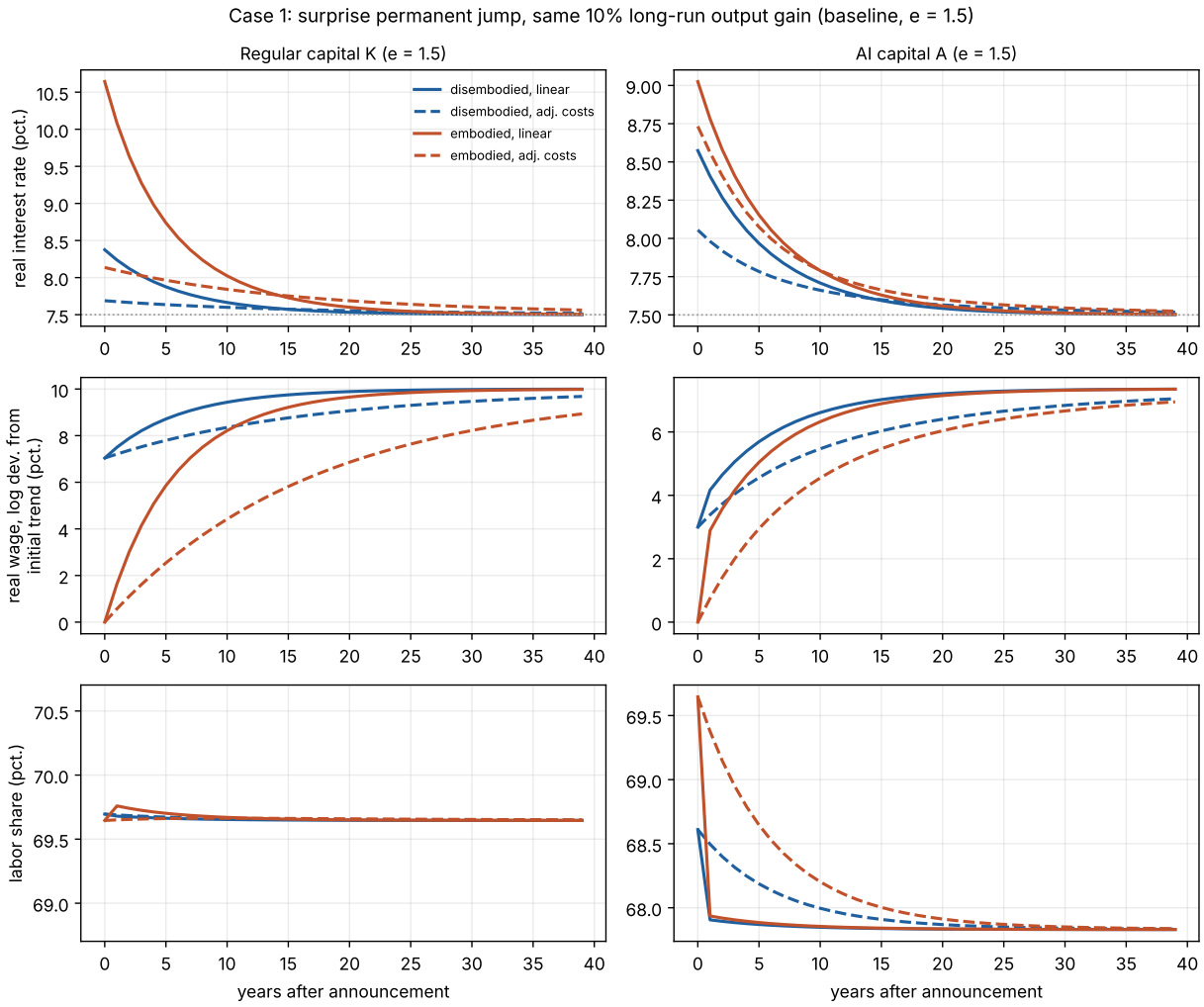


Figure 1: Case 1. Solid: linear accumulation. Dashed: adjustment costs.

Case 2: faster growth

Each shock raises long-run output growth by 0.5 percentage points, from 1.5 to 2.0 percent. AI growth needs $\varepsilon = 1$ (Proposition 3), so the figure and table use the $\varepsilon = 1$ variant for both capital stocks. Regular-capital results in the $\varepsilon = 1.5$ baseline are nearly identical; for example, r_1 under embodied progress with linear accumulation is 6.50 vs 6.48 percent.

1. *Long run.* The interest rate rises one-for-one with growth, from 7.50 to 8.03 percent, for every kind of progress. The wage grows faster, but its level relative to the old trend shifts down: by 1.5 percent for disembodied progress, 4.9 percent for embodied regular-capital progress, and 3.1 percent for embodied AI progress. The embodied shifts are larger because of obsolescence.

2. *Disembodied progress* raises r at once and part of the way, to 7.6 percent with linear accumulation or 7.9 percent with adjustment costs. The paths are identical for AI and regular capital (TFP equivalence).

3. *Embodied progress first lowers r.* With linear accumulation, r_1 falls to 6.5 percent for regular

Table 2: Balanced growth paths (Proposition 1). γ is the annual rate of technical progress; W/Z is relative to the initial path at the same date-0 technology levels. Rates in percent.

		γ	g	r	u_K	u_A	W/Z	labor share
$\varepsilon = 1.5$	Initial	0.00	1.50	7.50	14.10	27.50	1.000	69.6
	K disembodied, γ	1.26	2.00	8.03	14.63	28.03	0.985	69.7
	K embodied, γ	1.26	2.00	8.03	15.99	28.03	0.951	69.7
$\varepsilon = 1$	Initial	0.00	1.50	7.50	14.10	27.50	1.000	69.6
	K disembodied	1.26	2.00	8.03	14.63	28.03	0.985	69.6
	K embodied	1.26	2.00	8.03	15.99	28.03	0.951	69.6
	AI disembodied	12.08	2.00	8.03	14.63	28.03	0.985	69.6
	AI embodied	12.08	2.00	8.03	14.63	41.08	0.969	69.6

Table 3: Case 2: permanent increase in the growth of technology from $t = 0$ ($\varepsilon = 1$).

		γ	real interest rate (pct.)				$\ln W$ vs. old trend (pct.)		W/Z level
		(pct./yr)	r_1	t=5	t=20	new BGP	t=10	t=30	new/old BGP
K disemb.	linear	1.26	7.58	7.80	8.01	8.03	3.7	13.2	0.985
K disemb.	adj. costs	1.26	7.94	7.96	8.00	8.03	4.2	13.5	0.985
K emb.	linear	1.26	6.48	7.24	7.98	8.03	0.8	9.7	0.951
K emb.	adj. costs	1.26	7.76	7.81	7.94	8.03	2.7	10.6	0.951
AI disemb.	linear	12.08	7.58	7.80	8.01	8.03	3.7	13.2	0.985
AI disemb.	adj. costs	12.08	7.94	7.96	8.00	8.03	4.2	13.5	0.985
AI emb.	linear	12.08	7.28	7.64	8.00	8.03	2.2	11.6	0.969
AI emb.	adj. costs	12.08	7.83	7.88	7.99	8.03	3.0	11.9	0.969

capital and 7.3 percent for AI. Capital keeps getting cheaper, so firms postpone investment, consumption rises at once, and expected consumption growth falls. The real wage dips slightly in the first few years. Adjustment costs make postponing costly and largely remove the effect (r_1 of 7.8 percent). The AI effect is smaller because AI is a small share of capital spending.

Case 3: anticipated S-curve innovation

Table 4: Case 3: anticipated S-curve. Columns as in Table 1.

		shock	r_1 (pct.)	peak r (pct.)	year	$\ln W$, t=10 (pct.)	$\ln W$, long run (pct.)	labor share long run (pct.)
K disemb.	linear	0.255	7.53	8.16	12	3.3	10.0	69.6
K disemb.	adj. costs	0.255	7.57	8.34	10	3.7	10.0	69.6
K emb.	linear	0.255	7.37	8.57	17	-0.6	10.0	69.6
K emb.	adj. costs	0.255	7.50	7.90	16	0.6	10.0	69.6
AI disemb.	linear	1.223	7.51	8.07	13	1.9	7.4	67.8
AI disemb.	adj. costs	1.223	7.53	8.15	11	2.1	7.4	67.8
AI emb.	linear	1.223	7.47	8.10	16	0.6	7.4	67.8
AI emb.	adj. costs	1.223	7.50	7.99	15	0.8	7.4	67.8

1. *Disembodied progress* raises r gradually ahead of arrival. It peaks at 8.2–8.3 percent for regular capital and 8.1–8.15 percent for AI, around years 10–13. The wage rises with the S-curve.

2. *Embodied regular-capital progress with linear accumulation* produces a two-phase response. Before the innovation arrives, firms let capital run down while they wait for cheaper capital.

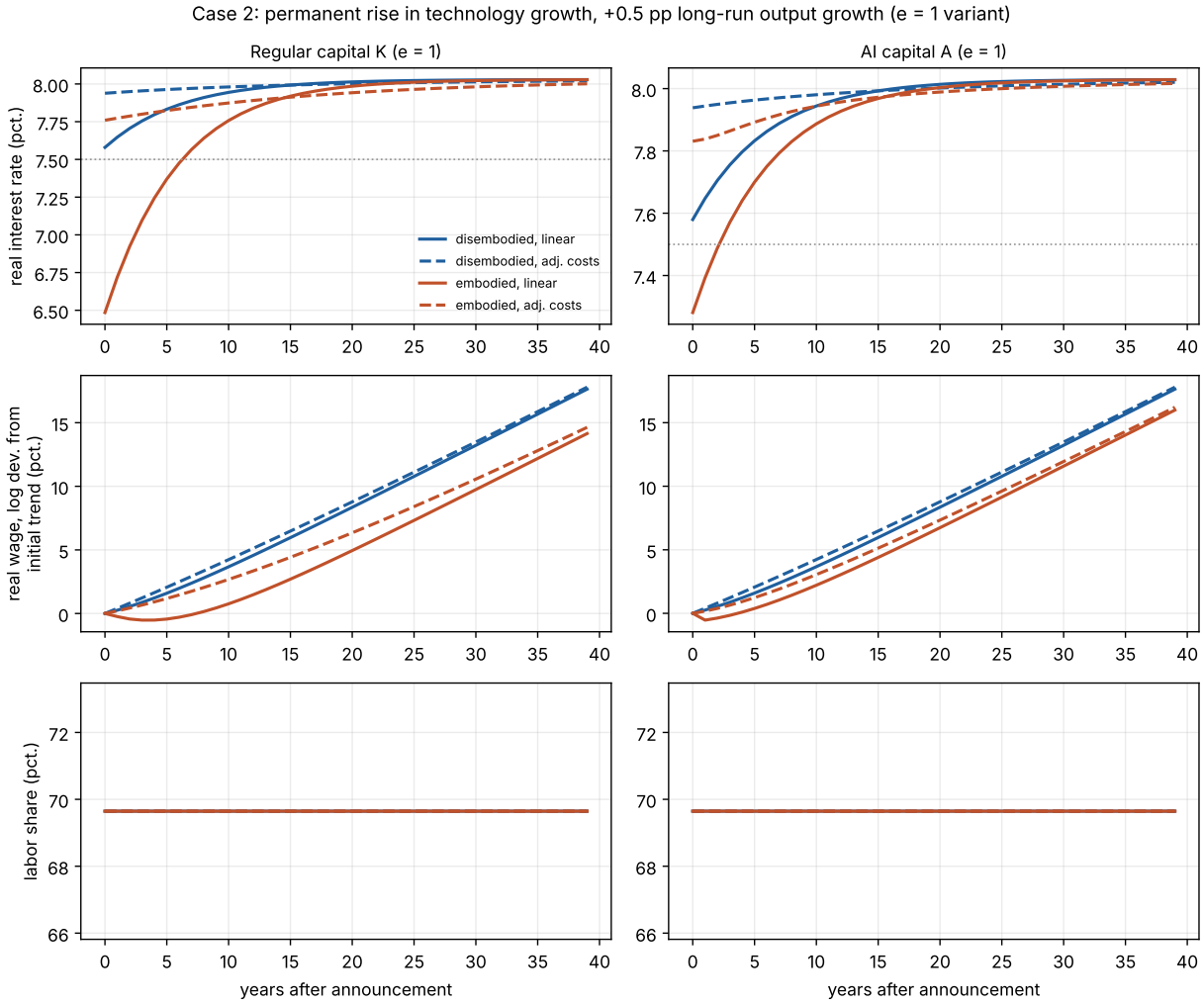


Figure 2: Case 2 ($\varepsilon = 1$). The labor share is constant by construction when $\varepsilon = 1$.

The real rate *falls* to about 6.8 percent around year 8 and the wage dips by about 1 percent. Then an investment boom pushes r to 8.6 percent in year 17. With adjustment costs, investment is spread out and r rises smoothly to 7.9 percent.

3. *Embodied AI progress* shows the same pattern in muted form, with a small dip before arrival and a peak of 8.0–8.1 percent around years 15–16.

4. *AI versus regular capital.* For the same output gain, AI progress produces similar or slightly smaller interest-rate peaks. It produces smaller wage gains (7.4 vs 10 percent) and a 1.8-point fall in the labor share that tracks the S-curve.

3 Summary

- *Wages.* For the same output gain, AI progress raises the wage by a fraction $1/[1+(1-a)(\varepsilon-1)]$ of the output gain, and regular-capital progress raises it one-for-one. The difference shows

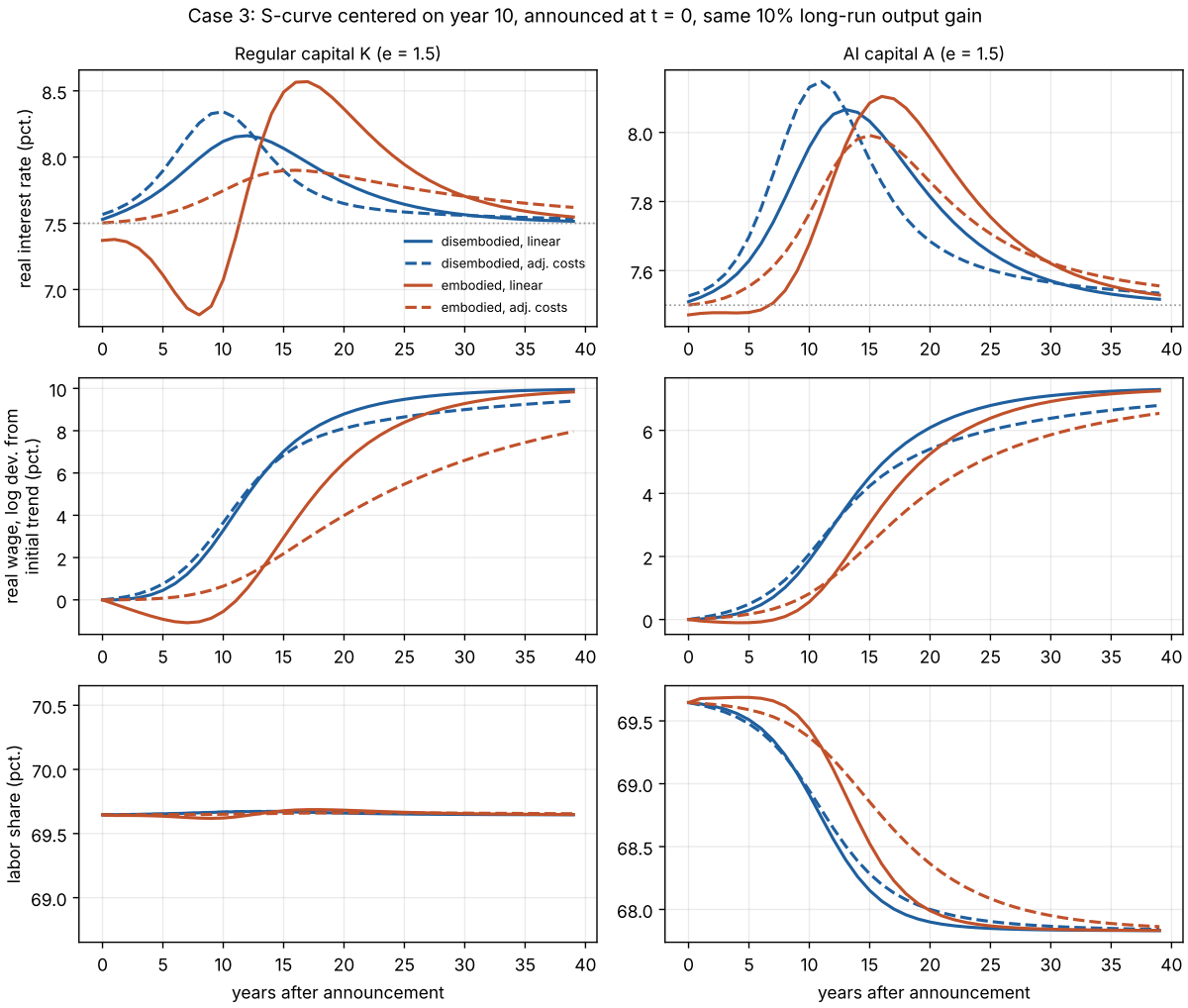


Figure 3: Case 3: S-curve centered on year 10, known at $t = 0$, same 10% long-run output gain.

up as a lower labor share.

- *Interest rates after jumps and announcements.* Embodied progress, whether in AI or regular capital, requires an investment boom and raises r more on impact than disembodied progress. Adjustment costs damp both responses.
- *Ongoing or anticipated embodied progress* can lower r at first, because firms wait for cheaper capital. The effect is strong with linear accumulation and largely absent with adjustment costs. It is weaker for AI than for regular capital as long as AI is a small share of capital.
- *Faster growth* raises r one-for-one in the long run for every kind of progress. With AI a substitute, permanently faster AI progress has no steady state. The AI share and the interest rate keep rising.

A Verification

`growth_cases.py` solves every experiment. It checks that the closed-form balanced growth path satisfies the equilibrium conditions, that long-run output gains match the target, and that the simulated growth rate converges to the closed form in case 2.

Check	max abs. error
e=1.5 adj=False: closed-form BGP satisfies equilibrium	1.3e-15
e=1.5 adj=True: closed-form BGP satisfies equilibrium	1.3e-15
e=1.0 adj=False: closed-form BGP satisfies equilibrium	7.8e-16
e=1.0 adj=True: closed-form BGP satisfies equilibrium	7.8e-16
level e=1.5 K dis lin: Newton residual	1.1e-12
level e=1.5 K dis lin: long-run ln Y gain vs target	3.0e-12
level e=1.5 K dis AC: Newton residual	2.7e-12
level e=1.5 K dis AC: long-run ln Y gain vs target	3.4e-08
level e=1.5 K emb lin: Newton residual	4.9e-15
level e=1.5 K emb lin: long-run ln Y gain vs target	1.3e-15
level e=1.5 K emb AC: Newton residual	8.5e-13
level e=1.5 K emb AC: long-run ln Y gain vs target	1.1e-07
level e=1.5 A dis lin: Newton residual	2.3e-14
level e=1.5 A dis lin: long-run ln Y gain vs target	5.4e-15
level e=1.5 A dis AC: Newton residual	3.0e-15
level e=1.5 A dis AC: long-run ln Y gain vs target	3.5e-08
level e=1.5 A emb lin: Newton residual	2.2e-15
level e=1.5 A emb lin: long-run ln Y gain vs target	1.3e-15
level e=1.5 A emb AC: Newton residual	3.8e-13
level e=1.5 A emb AC: long-run ln Y gain vs target	4.5e-08
growth e=1.5 K dis lin: Newton residual	1.0e-14
growth e=1.5 K dis lin: growth at T vs closed form	2.2e-16
growth e=1.5 K dis AC: Newton residual	1.4e-12
growth e=1.5 K dis AC: growth at T vs closed form	8.7e-11
growth e=1.5 K emb lin: Newton residual	2.3e-15
growth e=1.5 K emb lin: growth at T vs closed form	0.0e+00
growth e=1.5 K emb AC: Newton residual	3.6e-14
growth e=1.5 K emb AC: growth at T vs closed form	1.5e-10
growth e=1.0 K dis lin: Newton residual	1.0e-14
growth e=1.0 K dis lin: growth at T vs closed form	4.4e-16
growth e=1.0 K dis AC: Newton residual	1.4e-12
growth e=1.0 K dis AC: growth at T vs closed form	8.6e-11
growth e=1.0 K emb lin: Newton residual	1.1e-15
growth e=1.0 K emb lin: growth at T vs closed form	2.2e-16
growth e=1.0 K emb AC: Newton residual	3.7e-14
growth e=1.0 K emb AC: growth at T vs closed form	1.5e-10
growth e=1.0 A dis lin: Newton residual	1.0e-14
growth e=1.0 A dis lin: growth at T vs closed form	2.2e-16
growth e=1.0 A dis AC: Newton residual	1.4e-12
growth e=1.0 A dis AC: growth at T vs closed form	8.6e-11
growth e=1.0 A emb lin: Newton residual	1.2e-11
growth e=1.0 A emb lin: growth at T vs closed form	1.1e-15
growth e=1.0 A emb AC: Newton residual	1.1e-13
growth e=1.0 A emb AC: growth at T vs closed form	1.2e-10
scurve e=1.5 K dis lin: Newton residual	9.4e-13
scurve e=1.5 K dis lin: long-run ln Y gain vs target	2.5e-12
scurve e=1.5 K dis AC: Newton residual	3.9e-12
scurve e=1.5 K dis AC: long-run ln Y gain vs target	6.3e-08
scurve e=1.5 K emb lin: Newton residual	4.1e-15
scurve e=1.5 K emb lin: long-run ln Y gain vs target	9.3e-16
scurve e=1.5 K emb AC: Newton residual	2.9e-13
scurve e=1.5 K emb AC: long-run ln Y gain vs target	2.1e-07
scurve e=1.5 A dis lin: Newton residual	1.9e-11
scurve e=1.5 A dis lin: long-run ln Y gain vs target	8.8e-12
scurve e=1.5 A dis AC: Newton residual	3.4e-11
scurve e=1.5 A dis AC: long-run ln Y gain vs target	6.2e-08
scurve e=1.5 A emb lin: Newton residual	1.8e-15
scurve e=1.5 A emb lin: long-run ln Y gain vs target	6.9e-17
scurve e=1.5 A emb AC: Newton residual	2.1e-15
scurve e=1.5 A emb AC: long-run ln Y gain vs target	8.6e-08

References

Kiley, M. T. (2000). "Stock Prices and Fundamentals in a Production Economy." FEDS Working Paper 2000-05, Federal Reserve Board.

Kiley, M. T. (2004). "Stock Prices and Fundamentals: A Macroeconomic Perspective." *Journal of Business* 77(4): 909–936.